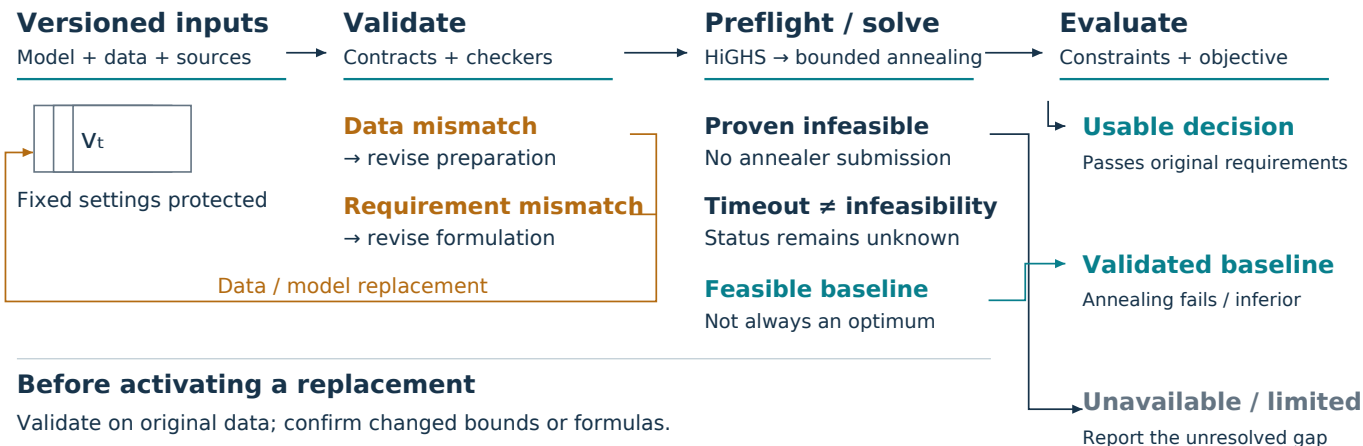

Before activating a replacement

Validate on original data; confirm changed bounds or formulas.
Retain earlier usable version; tie every solution to its model/data version.
Finite repair attempts, solver calls and process timeouts.

Solver failure alone does not establish a model error; penalty tuning remains a bounded inner loop.